

Table 2
Tests for sorting.

	(1) Chem 1A GPA	(2) Prior Cum. GPA	(3) Year in College	(4) EOP	(5) Female
URM * URM TA	0.1026 (0.0746)	−0.0142 (0.0328)	−0.0071 (0.0750)	0.0442 (0.0534)	0.0016 (0.0612)
Observations	836	837	837	837	837
R-squared	0.6746	0.6458	0.7486	0.7318	0.5529
Dep Var Mean	2.8018	3.1726	1.9663	0.4055	0.6045
Dep Var SD	0.5501	0.2300	0.4150	0.2622	0.2230

Notes: This table displays results from regressions on the minority-specific average student outcomes in a lab-section on an indicator equal to one if the average is associated with minority students, an indicator if the section is taught by a minority TA, the interaction between these two variables, and lab-section fixed effects. We only report the interaction term, to be interpreted as the extent that minority students sort into sections taught by minority TAs. Each column represents a regression on section averages by URM status. Underrepresented minorities include Latinx, African-American, American Indian / Alaska Native, and Native Hawaiian / Other Pacific Islander. Standard errors are clustered by teaching assistant. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3
Regression of Underrepresented Minority Student Outcomes on Assignment to Underrepresented Minority Teaching Assistants.

	(1) Dropped Course	(2) Passed Course	(3) Total Numeric Score(Standardized)	(4) Grade 4-Point Scale(Standardized)
URM*URM TA	−0.0548*** (0.0175)	0.0475** (0.0205)	−0.0796 (0.0865)	−0.0248 (0.0796)
Observations	4288	4288	4016	4022
R-squared	0.1538	0.1553	0.3007	0.2077
Dep Var Mean	0.0602	0.9363	0.0000	0.0000
Dep Var SD	0.2378	0.2442	1.0000	1.0000

Notes: Controls include lab-section fixed effects, and dummy variables for the full Chemistry 1A grade distribution, race/ethnicity, gender, Educational Opportunity Program (EOP) status, year in college, the most common majors (proposed or declared), and undeclared major. Underrepresented minorities include Latinx, African-American, American Indian / Alaska Native, and Native Hawaiian / Other Pacific Islander. Standard errors are clustered by teaching assistant. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results using several different student background variables are presented in Table 2. Standard errors are clustered at the TA level. We use the following five outcome variables, corresponding to the variable $\bar{X}_{mc}$ in Eq. (3.2): prior Chemistry 1A grade, prior cumulative grade point average, student's year in college, EOP status, and gender. Chemistry 1A is the first lecture course taken in the introductory sequence and is taken in a prior term to enrollment in the introductory labs. As noted below it is a strong predictor of course outcomes and is a particularly good measure of a potential unobserved student component that might be related to differential selection. We also have a measure of the student's cumulative GPA prior to enrolment in the first lab. As past GPA and current grades are highly correlated, this variable is also very useful for the sorting test. If the minority-non-minority gap of previous grades prior to enrolment in the current course is different in lab sections that are taught by minority instructors, our assumption of no differential sorting is most likely violated.

We do not find evidence of differential sorting: None of the estimates are statistically significant at any conventional level. Most importantly, minority gaps in accumulated GPA prior to course enrolment do not depend on TA race. In other words, we do not find evidence that high-ability minority students are more likely to take minority-taught labs compared with high-ability non-minority students.

Unlike differential sorting, absolute sorting is not a threat to validity for our identification. Nonetheless, we strongly believe that it is infeasible for students to select their TAs and that this form of absolute sorting is testable by examining correlations between TA race and student characteristics. We test this hypothesis by regressing an indicator of TA race or minority status on student characteristics. These results are reported on Table A1 and confirm that there is no evidence that students are sorting. Similarly, we replicate the falsification test outlined by Lusher et al. (2018) to see whether class average characteristics can predict TA assignment and we find no evidence of sorting (as reported in Table A2). Separately, we also test to see if the URM status of the first

assigned TA can predict future assignments, and we find no sorting in this dimension. We interpret the results from these tests as strong evidence in favor of our working hypothesis of no sorting, which is consistent with the TA assignment process.

4. Results

4.1. Main results

To examine the effects of minority student and graduate student teaching assistant interactions on course performance in the chemistry labs, we estimate Eq. (3.1). Estimates are reported in Table 3 for the main course outcomes. Specification 1 reports coefficient estimates for the minority student/minority TA interaction for the dependent variable – whether the student dropped the course.¹⁹ Minority students are substantially less likely to drop chemistry labs when assigned to a minority TA. We find an interaction coefficient of −0.055 which is statistically significant and large relative to the base drop level of 0.060. We also find a large positive effect on passing the course unconditionally. Estimates of the URM student - URM TA interaction indicate that passing the course increases by 0.048 on a base of 0.936. Most of the effect is driven by drops but some students do not pass the course conditional on enrolment. Passing the lab course is essential for completing the required introductory sequence in chemistry and might lead to disruptions in continuation (which we examine below).

We turn to performance in the lab course conditional on receiving a score or grade in the system. Specification 3 reports estimates for the

¹⁹ Drops here are defined as dropping the lab course that term independent of the section. The results are very similar if we measure dropped sections instead. Because of the restrictions noted above it is extremely difficult for a student to drop a lab course section and switch to a different one that term.